Tuple Coprimality and The First Hardy-Littlewood Conjecture

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Abstract

We encode coprimality information with respect to tuples in extensions of common arithmetic functions, and explore the relationship between these extensions and the First Hardy-Littlewood Conjecture. This leads to the appearance of the singular series of that conjecture in another context.

1 Tuples

The problems discussed are traditionally stated in terms of tuples, though they are entirely ill-suited to the task. Tuples are ordered, which is not required, they allow repeated elements, which we wish to forbid, and they are finite, which is not always necessary. As such, we use the following non-standard definition.

Definition 1.1. *Define* $\mathbb{K}$*, the set of* ***tuples****, by*

$$\mathbb{K} := \mathscr{P}(\mathbb{Z}).$$

If we require finiteness, we will indicate it manually. We denote the size of a set A by $\#A$. Our first goal is to view traditional number theoretic results as statements on the tuple $\{0\}$, and to extend those results in a natural way to all tuples.

Definition 1.2. *For* $A, B \in \mathbb{K}$*, we define*

$$A + B := \{a + b : a \in A, b \in B\}.$$

For $j \in \mathbb{Z}$ and $A \in \mathbb{K}$, define for convenience $j + A := \{j\} + A$. That is, $j + A$ is the translation of A by j.

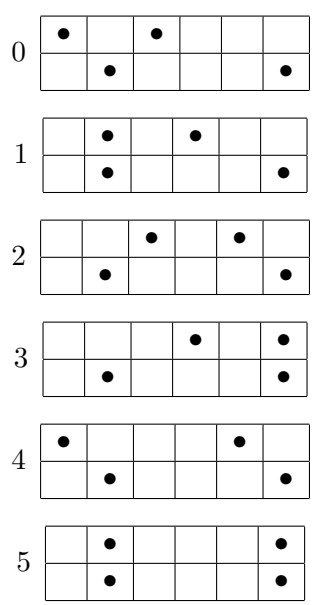
We are concerned with congruence between elements of tuples, but the usual definition of congruence is insufficient, as it describes only exact incidence between numbers relative to a modulus. We are instead concerned with a variety of types of incidence between translated tuples and the integers coprime to some modulus. The following definition handles the "modulus" aspect of that need.

Definition 1.3. *Given a tuple A and a natural number n, define*

$$A_n := \{a \bmod n : a \in A\}$$

2 The Totient Function

We now extend Euler's totient function φ to tuples. Instead of counting which natural numbers up to n are coprime to n, we count the translates of a tuple A by $j \in \mathbb{Z}_n$ for which all elements are coprime to n. We may also consider counting many other types of occurences, and will do so in a later section. We graph the counting process below for $A = \{0, 2\}$ and $n = 6$. The top grid represents a translate of A, and the bottom are those integers up to 6 coprime to 6. Recalling that those coprime integers are periodic with period 6, we simply wrap any elements of the translated A which fall off the end back to 0.



There is only a single occurrence, on table 5, of complete containment, so our totient should return 1 for this A and n.

Definition 2.1 (Tuple totient)**.**
For a tuple A, define φ_A by:

$$\begin{aligned}
&\varphi_A(1) := 1 \\
&\varphi_A(n) := \#\{j \in \mathbb{Z}_n : \forall\ a \in A,\ (j+a, n) = 1\}
\end{aligned}$$

Theorem 2.2. *For any tuple A, φ_A is multiplicative.*

Proof. Throughout, for any integer j, let j_n be the class represented by j in $\mathbb{Z}_n$. Let A be a tuple, and let m and n be coprime natural numbers. For each natural number b, define the set $P(b)$ by

$$P(b) = \{j_b \in \mathbb{Z}_b : \forall\ a \in A,\ (j_b + a, b) = 1\}.$$

Then by definition $\#P(a) = \varphi_A(a)$.

Now let $\Gamma : P(mn) \longmapsto P(m) \times P(n)$ be given by

$$\Gamma(j_{mn}) = (j_m, j_n).$$

We will show each of the following:

1. Γ **is well defined.** Let $j_{mn} \in P(mn)$. Then $\forall\ a \in A,\ (j+a, mn) = 1$, so $\forall\ a \in A,\ (j+a, m) = 1$. Proceeding likewise for n, $(j_m, j_n) \in P(m) \times P(n)$, so Γ is well defined.

2. Γ **is injective.** Let $a_{mn}, b_{mn} \in P(mn)$ such that $\Gamma(a_{mn}) = \Gamma(b_{mn})$. Then $a_m = b_m$ and $a_n = b_n$. Because m and n are coprime, $a_{mn} = b_{mn}$. Thus, Γ is injective.

3. Γ **is surjective.** Let $(b_m, c_n) \in P(m) \times P(n)$. By the Chinese Remainder Theorem, there exists $d_{mn} \in \mathbb{Z}_{mn}$ such that $d_m = b_m$ and $d_n = c_n$. It then suffices to show $d_{mn} \in P(mn)$, for this gives $\Gamma(d_{mn}) = (b_m, c_n)$.

   Let $a \in A$. We have

   $$\begin{aligned}
   &b_m \in P(m) \\
   \implies &(a + b_m, m) = 1 \\
   \implies &(a + d_m, m) = 1 \\
   \implies &(a + d_{mn}, m) = 1.
   \end{aligned}$$

   Proceeding likewise for c_n, we have $(a + d_{mn}, n) = 1$. Combining these gives $(a + d_{mn}, mn) = 1$, so $d_{mn} \in P(mn)$ and Γ is surjective.

Thus, Γ is a bijection, and

$$\begin{aligned}\varphi_A(mn) &= \#P(mn) \\ &= \#(P(m) \times P(n)) \\ &= \#P(m) \cdot \#P(n) \\ &= \varphi_A(m) \cdot \varphi_A(n).\end{aligned}$$

□

3 The Unit Function

The singular unit function $u(n)$ is defined by $u(n) = 1$ for all n. It is surprising that such a trivial function has a non-trivial extension to tuples.

Definition 3.1. *For any tuple A and prime p, define $\nu_A(p)$ by:*

$$\nu_A(p) = \#A_p$$

We extend ν_A to all natural numbers by complete multiplicativity.

In some literature, $\nu_A(p)$ is denoted $\nu_p(A)$, though it is usually defined only for primes.

Theorem 3.2. *For any tuple A, prime p, and natural number k we have:*

$$\varphi_A(p^k) = p^k - \nu_A(p)p^{k-1}.$$

Proof. By definition we have

$$\begin{aligned}\varphi_A(p^k) &= \#\{j \in \mathbb{Z}_{p^k} : \forall\ a \in A,\ (j+a, p^k) = 1\} \\ &= \#\{j \in \mathbb{Z}_{p^k} : \forall\ a \in A_{p^k},\ (j+a, p^k) = 1\}.\end{aligned}$$

If this condition is satisfied by some j, it will be satisfied by $j + pm$ for any m, and there are p^{k-1} multiples of p in $\mathbb{Z}_{p^k}$. In other words, as A_{p^k} is translated through $\mathbb{Z}_{p^k}$, we gain the same amount towards our total count as we pass through each period of size p, and there are p^{k-1} such periods. We then have

$$\begin{aligned}\varphi_A(p^k) &= p^{k-1}\#\{j \in \mathbb{Z}_p : \forall\ a \in A_p,\ (j+a, p) = 1\} \\ &= p^{k-1}(p - \#\{j \in \mathbb{Z}_p : \exists\ a \in A_p,\ (j+a, p) \neq 1\})\end{aligned}$$

Now, $(j+a,p) \neq 1$ exactly once per $a \in A_p$ as j ranges over $\mathbb{Z}_p$ (when $j+a \equiv 0 \bmod p$). Thus, the size of the set in the previous equation is $\#A_p$, which gives the required formula. □

Theorem 3.3 (Totient Product Formula)**.**
For any tuple A and natural number n, we have:

$$\varphi_A(n) = n \prod_{p|n} \left(1 - \frac{\nu_A(p)}{p}\right).$$

Proof. Let A be a tuple and n be a natural number. Then n has a factorization, say

$$n = \prod_{p_i|n} p_i^{a_i}.$$

Then we have by theorems 2.2 and 3.2,

$$\begin{aligned}
\varphi_A(n) &= \varphi_A\left(\prod_{p_i|n} p_i^{a_i}\right) \\
&= \prod_{p_i|n} \varphi_A(p_i^{a_i}) \\
&= \prod_{p_i|n} \left(p_i^{a_i} - \nu_A(p_i) p_i^{a_i-1}\right) \\
&= n \prod_{p_i|n} \left(1 - \frac{\nu_A(p_i)}{p_i}\right).
\end{aligned}$$

□

Theorem 3.4. *For any tuple A, the φ_A has the following properties:*

(a) $\varphi_A(mn) = \varphi_A(m)\varphi_A(n)(d/\varphi_A(d))$, *where* $d = (m,n)$.

(b) $\varphi_A(mn) = \varphi_A(m)\varphi_A(n)$ *if* $(m,n) = 1$.

(c) $a|b$ *implies* $\varphi_A(a)|\varphi_A(b)$.

Proof. The proof of these follows nearly verbatim the proofs for φ, which are shown in Theorem 2.5 of [1]. □

Definition 3.5. *Let w be an arithmetic function such that:*

1. *w is completely multiplicative.*

2. *w is eventually constant, say $w(p) = k$ for $p > N$.*

3. *For all primes p, $w(p) \leq p$ and $w(p) \leq k$.*

*Then we say w is **unit looking**. If there exists a tuple such that $\nu_A = w$, we say A generates w, and say w is good. Otherwise, w is bad.*

Unit looking functions, as far as can be checked by simple methods, appear to have been generated by some tuple. For $k = 2$, all unit looking functions are good. Proving any individual unit looking function is bad is a tedious affair involving many cases, and so is omitted. A general algorithm determining this property would be useful in later computation, but it seems unlikely such an algorithm is feasable in the general case. As a concrete example, the unit looking function w with $w(2) = 1$, $w(3) = w(103) = 2$, and $w(p) = 3$ for all other primes p is bad, and is the minimal example. Computation shows that bad unit looking functions become far more common as the primes at which they differ from their eventual value grow, and it seems likely that the proportion of good functions goes to zero in that limit. We may, however, easily find tuples whose unit function matches a given unit looking function up to an arbitrary bound.

Theorem 3.6. *Let p_n be the nth prime number. Let w be a unit looking function. Then for any natural N, there exists a tuple A such that $\nu_A(n) = w(n)$ for $n \leq N$.*

Proof. By the complete multiplicativity of ν_A and w, we need only show the equality on primes up to N.

Let k be the eventual value of w on primes. Let $A_1 = \{0\}$. If $k = 1$, we have $\nu_{A_1} = w$ and are done. Now for each i from 2 to k, let Q_i be the set of primes up to N for which $\nu_{A_i}(p) < w(p)$. By the Chinese Remainder Theorem there exists a which solves $a \equiv i \pmod{q}$ for all $q \in Q_i$, and $a \equiv 0$ modulo each other prime up to N. Let $A_i = A_{i-1} \cup a$.

Then for each prime p_n up to N, exactly the first $w(p_n)$ residues have been filled by elements of A_k. That is, $\nu_{A_k}(p_n) = w(p_n)$. $\square$

4 Dirichlet Convolution

For any arithmetic functions f and g we denote their Dirichlet convolution by

$$(f * g)(n) := \sum_{ab=n} f(a)g(b).$$

We denote the Dirichlet inverse of an arithmetic function f by f^{-1}. We define the arithmetic function N by $N(n) = n$.

Definition 4.1. *For any tuple A, define μ_A to be the Dirichlet inverse of ν_A.*

By definition ν_A is completely multiplicative, so by Theorem 2.17 in [1], we have its Dirichlet inverse to be

$$\mu_A(n) = \nu_A^{-1}(n) = \mu(n)\nu_A(n).$$

We have also from Theorem 2.18 of [1] that for any multiplicative function f and natural number n, that

$$\sum_{d|n} \mu(d)f(d) = \prod_{p|n}(1 - f(p)), \tag{1}$$

which is used to prove the following.

Theorem 4.2. *For any tuple A, we have*

$$\nu_A * \varphi_A = N.$$

Proof. Applying equation (1) with $f(n) = \nu_A(n)/n$ to the totient product formula, we have

$$\begin{aligned}
\varphi_A(n) &= n \prod_{p|n} \left(1 - \frac{\nu_A(p)}{p}\right) \\
&= n \sum_{d|n} \mu(d) \frac{\nu_A(d)}{d} \\
&= \sum_{d|n} \mu_A(d) \frac{n}{d} \\
&= (\mu_A * N)(n).
\end{aligned}$$

Taking the Dirichlet product of both sides by ν_A, we obtain the theorem. □

This result, while simple, is one reason for our interest in ν_A. Intuitively, φ_A should encode information about coprimality in the tuple sense. The function N is neutral in this regard, so we should expect $\nu_A = N * \varphi_A^{-1}$ to retain some aspect of that information while playing nicely with Dirichlet series.

5 Admissibility

Definition 5.1. *We say a tuple A is* ***admissible*** *if, for each prime p, we have*

$$\nu_A(p) < p.$$

Then a tuple is admissible if it does not fill all of the residue classes modulo any prime. This condition gives a simple bound on ν_A.

Proposition 5.2. *For any admissible tuple A, we have*

$$\nu_A(n) \leq \varphi(n).$$

Proof. Let A be an admissible tuple, and let n be a natural number with factorization

$$n = \prod_{p_i | n} p_i^{a_i}.$$

Then

$$\begin{aligned}
\nu_A(n) &= \nu_A\left(\prod_{p_i|n} p_i^{a_i}\right) \\
&= \prod_{p_i|n} \nu_A(p_i)^{a_i} \\
&\leq \prod_{p_i|n} (p_i - 1)^{a_i} \\
&\leq \prod_{p_i|n} (p_i - 1) \cdot p_i^{a_i - i} \\
&= \varphi(n).
\end{aligned}$$

□

This allows stating the qualitative version of the First Hardy-Littlewood Conjecture (given later).

Conjecture 5.3 (Prime K-Tuple Conjecture)**.** *For any finite, admissible tuple A, there exists an infinite tuple B such that $A + B \subset \mathbb{P}$.*

Admissibility is required, for if a tuple A was inadmissible by prime p, for $b \in B$ large enough that each element of $b + A$ is greater than p, we must have some element of A filling the 0 residue. That element is then divisible

by p, $b+A \not\subset \mathbb{P}$, and it cannot be for any b beyond that bound. Thus, there can be only finitely many integers at which $b+A \subset \mathbb{P}$ and no valid B exists.

The finiteness of A is also required. For example, let A be the set of all primorials. This fills at most i residues of each prime p_i and so is admissible. We cannot have negatives in B, for $2 \in A$ is the minimum prime. We cannot have 0 or 1 in B, because $A+B$ would contain 6 and 4 respectively. If B contains $b > 1$, b is divisible by some prime p, and the primorials will eventually be divisible by p. Then for some element $a \in A$ divisible by p, $p|a+b$, so $A+B \not\subset \mathbb{P}$. Thus, there are no integers b such that $b+A \subset \mathbb{P}$. Not only can B not be infinite in this case, it cannot contain any integer.

If the conjecture holds, there exist infinite tuples A with arbitrary large finite B such that $A+B \subset \mathbb{P}$. Indeed, given a finite admissible tuple B, we apply the conjecture and find such an A. The question of adding two infinite tuples and obtaining a prime tuple is handled in greater generality in [2], the results of which we state in less generality as the following theorem. It is often said that the Prime K-Tuple Conjecture feels intuitively true, so it is somewhat alarming that it holds as a consequence the following result, which seems intuitively false.

Theorem 5.4. *Assume the Prime K-Tuple Conjecture. Then there exist infinite tuples A and B such that $A+B \subset \mathbb{P}$.*

6 The First Hardy-Littlewood Conjecture

The Prime K-Tuple Conjecture asserts the existence of an infinite number of copies of a tuple among the primes. The First Hardy-Littlewood conjecture (HLC1) asserts the asymptotic frequency of these occurences.

Definition 6.1. *For any tuple A, define*

$$\pi_A(x) = \#\{n \leq x : n + A \subset \mathbb{P}\}.$$

Definition 6.2. *For $n > 0$, define*

$$\mathrm{Li}_n(x) = \int_2^\infty \frac{\mathrm{dt}}{\log^n(t)}.$$

We call this the nth (offset) logarithmic integral.

Conjecture 6.3 (First Hardy-Littlewood conjecture)**.** *Let A be an admissible tuple. Then*

$$\pi_A(x) \sim C_A \mathrm{Li}_{\#A}(x),$$

where

$$C_A = \prod_p \frac{1 - \nu_A(p)p^{-1}}{(1-p^{-1})^{\#A}}.$$

This conjecture can be understood intuitively. Consider a translation of a tuple A by a large value j. If we assume that prime numbers are distributed randomly with a distribution of $1/\log(n)$, and that the probabilities of each element of $A+j$ being prime are uncorrelated, we would guess the probability of all the elements of $A+j$ being prime to be

$$\prod_{n \in j+A} 1/\log(n).$$

For our large j, this is essentially $1/\log(n)^{\#A}$. Then we should naively expect $\pi_A(x) \sim \mathrm{Li}_{\#A}(x)$.

The presence of C_A, sometimes called the 'singular series' for A, is a correction factor for our assumption of non-correlation. To see this, consider $A = \{0, 2\}$, the twin prime case. Translating this by our large j, if $j+0$ is prime, we know that $j+0$ is odd. Then $j+2$ is also odd, which should double the chance that $j+2$ is prime when compared to a randomly selected value in this range. We have also that $j+0 \equiv 1$ or $2 \pmod 3$, so $j+2 \equiv 0$ or $1 \pmod 3$. If it were randomly selected, there was a $1/3$ chance that $j+2$ was a multiple of 3 regardless of the primality of $j+0$; we now see that the structure of A dictates the actual chance should be $1/2$. Correcting probabilities with respect to the structure of A for each prime in this way gives exactly C_A.

This conjecture implies that the exact structure of a tuple does not determine its asymptotic behavior, but rather that behavior is entirely decided by ν_A. For example, $\{0, 2\}$ and $\{0, 4\}$ produce the same ν_A. This yields an equivalence relation.

Definition 6.4. *Given tuples A and B, we say they are* ***residually equivalent*** *if $\nu_A = \nu_B$.*

The following asymptotic expansion is well known:

$$\mathrm{Li}_1(x) \sim \frac{x}{\log x} \sum_{k \geq 0} \frac{k!}{\log^k x}. \tag{2}$$

The simpler estimate $\frac{x}{\log x}$ given in the Prime Number Theorem (PNT) is then the truncation of this expansion to its first term. Then we should expect to analagously use the truncation of the asymptotic expansion of Li_n

in attempting to gain traction in HLC1. Integration by parts and a simple induction gives

$$\mathrm{Li}_n(x) \sim \frac{x}{(n-1)!\log x} \sum_{k \geq n-1} \frac{k!}{\log^k x}.$$

If HLC1 is correct, we might then attempt to show, following in the steps of the PNT, that

$$\pi_A(x) \sim \frac{C_A x}{\log^{\#A} x}.$$

As with the PNT, this can be translated into

$$\Psi_A(x) \sim C_A x,$$

where $\Psi_A(x) = \sum_{n \leq x} \prod_{a \in A} \Lambda(a+n)$, and Λ is the Von Mangoldt function, for which $\Lambda(p^k) = \log p$, and which is zero for other inputs. In the case of the PNT, summing over one copy of Λ allows the decomposition of a later sum into a double sum over primes and powers, but the product of Λs here forbids this: we must instead contend with a sum over the n for which each $a_i \in A$ gives $n + a_i = p_i^{k_i}$. That is, we require knowledge of the prime factorizations of integers which differ by constant amounts. This is the critical difficulty of the problem, and any solution in this direction must at some point contend with this seemingly impossible task.

7 The Dirichlet Series of ν_A

We denote the Dirichlet series of an arithmetic function f by

$$\mathcal{D}(f)(s) := \sum_{n \geq 1} \frac{f(n)}{n^s}.$$

We may factor $\mathcal{D}(\nu_A)$ as

$$\mathcal{D}(\nu_A)(s) = \zeta^{\#A}(s) \prod_p \frac{(1-p^{-s})^{\#A}}{1-\nu_A(p)p^{-s}}. \tag{3}$$

We refer to the final product as the modulator, denoted $N_A(s)$.

Proposition 7.1. *For any admissible tuple A, there exists $\epsilon > 0$ such that $N_A(s)$ converges for $\Re(s) > 1 - \epsilon$.*

Proof. Let A be an admissible tuple. Then $\nu_A(p) < p$, so the real parts of the solutions of $1 - \nu_A(p)p^{-s} = 0$ are all left of 1, and go to 0 as $p \to \infty$. Then there exists $\epsilon > 0$ such that all solutions of $1 - \nu_A(p)p^{-s} = 0$ lie to the left of $1 - \epsilon$. Fix $s = \sigma + it$ with $\sigma > 1/2$. Let

$$N_{A,r}(s) = \prod_{p>r} \frac{(1 - p^{-s})^{\#A}}{1 - \nu_A(p)p^{-s}}.$$

We will show the convergence of $N_{A,r}$ at s for a fixed r, from which the result follows, as the finitely many removed factors all converge for $\Re(s) > 1 - \epsilon$. We fix r to be large enough that both $\#Ar^{-\sigma} < 1/2$, and $\nu_A(p) = \#A$ for $p > r$. We then have

$$\log N_{A,r}(s) = \sum_{p>r} \left(\#A \log(1 - p^{-s}) - \log(1 - \#Ap^{-s}) \right).$$

Expanding the logarithms as power series gives

$$\begin{aligned} \log N_{A,r}(s) &= \sum_{p>r} \left(-\#A \sum_{k\geq 1} \frac{p^{-sk}}{k} + \sum_{k\geq 1} \frac{(\#Ap^{-s})^k}{k} \right) \\ &= \sum_{p>r} \sum_{k\geq 2} \left(\frac{(\#A)^k - \#A}{k} p^{-sk} \right). \end{aligned}$$

Note the cancellation of the $k = 1$ term. Now,

$$|\log N_{A,r}(s)| \leq \sum_{p>r} \sum_{k\geq 2} \left(\#Ap^{-\sigma} \right)^k.$$

For each $p > r$, by our bound on r, the k sum is a convergent geometric series. We have also for $p > r$ that $1 - \#Ap^{-\sigma} > 1/2$, so

$$\begin{aligned} |\log N_{A,r}(s)| &= \sum_{p>r} \frac{(\#Ap^{-\sigma})^2}{1 - \#Ap^{-\sigma}} \\ &\leq 2(\#A)^2 \sum_{p>r} p^{-2\sigma} \\ &\leq 2(\#A)^2 \sum_{n>r} n^{-2\sigma}. \end{aligned}$$

We have $\sigma > 1/2$, so this is a convergent p-series. Thus, $\log N_{A,r}$ converges absolutely, and exponentiating gives the result. □

We now consider C_A of HLC1 not as a constant, but as a function of s defined by $C_A(s) = 1/N_A(s)$. We continue to use C_A to mean $C_A(1)$. Reciprocating equation (3), we now have

$$\frac{1}{\mathcal{D}(\nu_A)(s)} = \frac{C_A(s)}{\zeta^{\#A}(s)}.$$

Comparing this to HLC1 is, at least intuitively, highly suggestive.

The Laurent expansion of ζ is known, so the residues above are computable. We then compute $N_A(1)$ for some admissible, non-residually-equivalent tuples, where the p column gives those primes over which $\nu_A(p) \neq \#A$. Computations were made using the product over the first 10000 primes before rounding, and are made only for the sake of discussion: they have not been verified for accuracy. We also include the corresponding C_As.

A	p	$N_A(1)$	C_A
$\{0,2\}$	2	0.75738949...	1.32032464...
$\{0,10\}$	2,5	0.56804211...	1.76043285...
$\{0,6\}$	2,3	0.37869474...	2.64064928...
$\{0,2,6,8\}$	2,3	0.24089421...	4.15119988...
$\{0,8,12,30\}$	2,3,5,11	0.10539121...	9.48845686...
$\{0,6,12,18\}$	2,3	0.12044710...	8.30239975...
$\{0,2,6,8,30\}$	2,3,5,7,11	0.05639911...	17.73077650...
$\{0,18,30,60\}$	2,3,5,7	0.03011178...	33.20959900...
$\{0,60,120,180\}$	2,3,5	0.03011178...	33.20959900...

We see that the final two entries appear to be exactly equal. It is not a rounding error: they are identical. This, along with their corresponding tuples having the same size, gives that the Hardy-Littlewood Conjecture predicts they will have identical limiting frequency, despite their residual inequivalence.

8 Alternatives to φ_A

When we defined φ_A, we counted occurences of complete coprimality. We could also ask for the occurences of a non-zero amount of coprimality, a specific amount, or a minimum amount, among other requirements. However, these examples do not produce multiplicative φ analogs.

There is also a subtlety in defining many of them that doesn't exist in our φ_A. We defined φ_A in terms of translates of A, but we could have equivalently counted the translates of A_n. These definitions are inequivalent

for the examples listed above, for we will have $\#A_n < \#A$ when elements of A coincide modulo n. That is, the issue with those examples is that they rely on the number of $a \in k + A$ or $a \in k + A_n$ which satisfy a property, and these may differ. We may avoid this issue by only considering functions which count those k such that each of the elements of $k + A$ possesses some property, rather considering a property on $k + A$ itself. But to consider a function to be an analog of φ, the property considered should be "Are these elements coprime?". There is only one function of this type which considers this property: φ_A.

We may relax our constraints, though, on what property constitutes a totient to depend on the value of $(k + a, n)$, rather than if it is 1 or not. This gives not only additional tuple totients, but also additional singular totients. In general, we may use the property that $(k + a, n)$ is limited to a specific subset of the primes and obtain a multiplicative totient.

Definition 8.1. *Given a set of primes B, let $a \prec B$ mean that a is composed only of elements of B.*

In particular, if this set is exactly the primes less than some bound b, we are requiring $(k + a, n)$ to be b-smooth:

Definition 8.2. *If the prime factors of an integer k are bounded by b, we say k is b-**smooth**.*

Definition 8.3. *For a tuple A and a set of primes B, define $\varphi_{A,B}$ by*

$$\varphi_{A,B}(1) := 1,$$
$$\varphi_{A,B}(n) := \#\{k \in \mathbb{Z}_n : \forall\ a \in A,\ (k + a, n) \prec B\}.$$

Theorem 8.4. *For any tuple A and set of primes B, $\varphi_{A,B}$ is multiplicative.*

Proof. The proof follows exactly that of φ_A and so is omitted. □

What, then, are the corresponding unit functions $\nu_{A,B}$? They are the same as our earlier ν_A, except they are zero at every number containing a factor in B. They are completely multiplicative, can be used to compute $\varphi_{A,B}$ in the same way, and yield $\nu_{A,B} * \varphi_{A,B} = N$. The proofs of these facts follow exactly those for ν_A. We return now to the realm of φ_A, keeping in mind that many results may be extended trivially in this way to $\varphi_{A,B}$.

The earlier nonmultiplicative examples are not without value, and not beyond hope. In essentially all cases, we can express the desired function as a linear combination of φ_A. Consider first for some n and tuple A, counting the $k \in \mathbb{Z}_n$ such that exactly q elements of $k + A$ are coprime to n. We call the value q the quota, and refer to this as the unwrapped, exact quota case.

Definition 8.5 (Quota Tuple Totients)**.** *For any tuple A and quota $q \in \mathbb{Z}$ with $0 \leq q \leq \#A$, define $\Upsilon_A^{q=}$ by*

$$\Upsilon_A^{q=}(1) := 1$$
$$\Upsilon_A^{q=}(n) := \#\{k \in \mathbb{Z}_n : q = \#\{a \in A : (k+a, n) = 1\}\}.$$

Replacing A with A_n in this definition yields a wrapped case, and requiring at least q values to be coprime to n yields a minimum quota case. We denote the wrapped cases with ϱ rather than Υ, and minimum quotas use $q \leq$ in the superscript. Clearly, $\Upsilon_A^{\#A=} = \varphi_A$, though this is also proven by the following theorem.

Theorem 8.6. *For any tuple A and quota q,*

$$\Upsilon_A^{q=} = \sum_{k=q}^{\#A} (-1)^{k-q} \binom{q}{k} \sum_{\substack{B \subset A \\ \#B = k}} \varphi_B$$

Proof. TODO □

Theorem 8.7. *For any tuple A, and quota q,*

$$\Upsilon_A^{q\leq} = \sum_{k=q}^{\#A} (-1)^{k-q} \binom{k-1}{k-q} \sum_{\substack{B \subset A \\ \#B = k}} \varphi_B.$$

Proof. TODO □

The formulae for the corresponding ϱ functions are identical: simply replace A with A_n throughout. This creates a dependance on n in the final sum which appears to explode the computational complexity, but for sufficiently large n, $A = A_n$, so the order of computational complexity is the same. This also shows that the wrapped and unwrapped totients are eventually equivalent, as should be expected. We will make no further use of the wrapped totients, and the proofs are mostly the same as the unwrapped cases, so they are omitted.

Using the discrete modular convolution, we can obtain the essential data required to compute wrapped quota totients for all quotas at once.

Definition 8.8. *Given $f, g : \mathbb{Z}_n \to \mathbb{C}$, define the discrete modular convolution $\odot_n$ by*

$$(f \odot_n g)(k) = \sum_{\substack{a+b \equiv k \\ \bmod n}} f(a)g(b),$$

where $a, b \in \mathbb{Z}_n$.

The symbol $\odot$ is non-standard. Normally $*$ would be used, but it is already in use for Dirichlet convolution.

Theorem 8.9. *Let A be a tuple, let $n \in \mathbb{N}$, and let χ_0 be the principle Dirichlet character mod n. Let 1_{A_n} be the indicator function of A_n. Then*

$$(1_{A_n} \odot_n \chi_0)(k) = \#\{a \in A_n : (k + a, n) = 1\}.$$

Proof. TODO $\square$

Computing $1_{A_n} \odot_n \chi_0$ for all k then gives a convenient data format for querying quota totient information at n. This can be made to work for unwrapped quota totients by considering a period of size n of the non-modular discrete convolution of χ_0 and 1_A.

9 TODO

1. Decide theorem vs proposition for each result.
2. Double check Granville actually implies the thing.
3. We should check if any results require finiteness on tuple. Also check each theorem for admissibility requirements.

References

[1] Tom M. Apostol. *Introduction to Analytic Number Theory*. Springer, 1976. ISBN: 9788184895216.

[2] Andrew Granville. *A note on the sums of primes*. Tech. rep. University of Toronto, 1990.